

heap built up over generations from the discarded shells of harvested mollusks. Junko Habu and colleagues' survey of Japanese shell midden archaeology documents thousands of such sites along the Pacific coast, with especially dense concentrations around bays like Matsushima, and traces intensive shellfish collecting back to roughly 8000–7000 BCE, when a radiocarbon-dated oyster shell from the Natsushima site marks the start of large-scale marine exploitation in the Early Jomon period (Habu et al. 2011). Shell middens preserve well and accumulate in seasonal layers, so they let archaeologists reconstruct not just which species people harvested but when. At the Awazu site, alternating bands of freshwater clam and nutshell record repeated spring, summer, and autumn collection episodes, and caloric reconstructions there found that nuts, not shellfish, supplied the largest share of the estimated diet: 52 percent, against 16.7 percent from shellfish. The Isarago midden tells a similar story, where simulation studies estimated that shellfish supplied only about 7 to 9 percent of daily caloric intake (Habu et al. 2011). Shellfish, in other words, mattered to Jomon communities less as a caloric staple than as insurance: a reliable resource during the late winter and early spring, when other foods ran thin, buffering risk rather than dominating the diet. Stable isotope analysis of human bone collagen, a method that measures the ratios of carbon and nitrogen isotopes retained in skeletal tissue to estimate how much of a person's protein came from marine versus terrestrial sources, reinforces this picture. Kusaka and colleagues' isotopic study of Jomon skeletons from two coastal regions found a mixed marine-terrestrial diet whose balance shifted with local resource availability, consistent with terrestrial foods dominating the overall Jomon diet even in coastal, shell-midden-forming communities (Kusaka et al. 2010). A related isotopic study of 123 individuals from the Yoshigo shell mound in Aichi Prefecture, dated to roughly 3200–2800 cal BP, found substantial variation in marine dietary dependence within a single Jomon population, correlated with burial cluster

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Defining the dates for this in parentheses would be helpful

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Good presentation of specific evidence here

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I would break this up and start a new paragraph here.